

L14



L14: Proof of Quadratic Reciprocity Law ^① (Theorem 4.9)

Lemma 4.10: Let p be an odd prime number and let $a \in \mathbb{Z}$ with $p \nmid a$ and a odd. Let

$$N := \sum_{j=1}^{(p-1)/2} \left[\frac{ja}{p} \right].$$

Then,

$$\left(\frac{a}{p} \right) = (-1)^N.$$

Here, $[\cdot]$ denotes the floor function.

• Lemma 4.10, similar to Euler's Criterion and Gauss' Lemma, is of more theoretical interest rather than computational.

Example: Use Lemma 4.10 to compute $\left(\frac{7}{11} \right)$.

We have

$$\begin{aligned} N &= \sum_{j=1}^5 \left[\frac{j \cdot 7}{11} \right] = \left[\frac{7}{11} \right] + \left[\frac{14}{11} \right] + \left[\frac{21}{11} \right] + \left[\frac{28}{11} \right] \\ &\quad + \left[\frac{35}{11} \right] \\ &= 0 + 1 + 1 + 2 + 3 = 7. \end{aligned}$$

Thus $\left(\frac{2}{11}\right) = (-1)^7 = -1.$ (2)

Proof (of Lemma 4.10): Let $r_1, \dots, r_n$ be the least non-negative residues of the integers $a, 2a, \dots, \left(\frac{p-1}{2}\right)a$ that are greater than $p/2$, and let $s_1, s_2, \dots, s_m$ be the least non-negative residues of these integers that are less than $p/2$. Then for $j = 1, 2, \dots, \frac{p-1}{2}$, we have that

$$\textcircled{1} \quad ja = p \left[\frac{ja}{p} \right] + \text{remainder depending on } j.$$

Each of the $r_1, r_2, \dots, r_n, s_1, s_2, \dots, s_m$ appears exactly once as a remainder.

By adding the $\frac{p-1}{2}$ equations in $\textcircled{1}$, we obtain

$$\sum_{j=1}^{(p-1)/2} ja = \sum_{j=1}^{(p-1)/2} p \left[\frac{ja}{p} \right] + \sum_{j=1}^n r_j + \sum_{j=1}^m s_j. \quad \textcircled{2}$$

From the proof of Gauss' Lemma, we

know that the integers $p-r_1, p-r_2, \dots, p-r_n, s_1, s_2, \dots, s_m$ are precisely the integers from 1 to $\frac{p-1}{2}$ inclusive in some order. Thus

$$\sum_{j=1}^{(p-1)/2} j = \sum_{j=1}^n (p-r_j) + \sum_{j=1}^m s_j = pn - \sum_{j=1}^n r_j + \sum_{j=1}^m s_j. \quad (3)$$

Subtracting (3) from (2) we obtain

$$\sum_{j=1}^{(p-1)/2} ja - \sum_{j=1}^{(p-1)/2} j = \sum_{j=1}^{(p-1)/2} p \left[\frac{ja}{p} \right] - pn + 2 \sum_{j=1}^n r_j. \quad (4)$$

Reducing both sides of (4) modulo 2, we have

$$0 \equiv \sum_{j=1}^{(p-1)/2} \left[\frac{ja}{p} \right] - n \pmod{2},$$

and so

$$n \equiv \sum_{j=1}^{(p-1)/2} \left[\frac{ja}{p} \right] \pmod{2}.$$

Thus $n \equiv N \pmod{2}$, and so $\left(\frac{a}{p} \right) = (-1)^n$ (by Gauss Lemma) implies that $\left(\frac{a}{p} \right) = (-1)^N. \quad \square$

Proof (of Theorem 4.9 - Quadratic reciprocity): ④

We will prove that $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$.

Without loss of generality we assume that $p > q$. Consider Figure 4.1 on pg 122 of the textbook. We will count, in two different ways, the number of lattice points in the rectangle $OABC$, including those points on AB and BC , but not those on OA and OC . This number is clearly $\frac{(p-1)}{2} \cdot \frac{(q-1)}{2}$.

We will count a second way now. We have the following facts:

(a) ON has slope q/p . Since p and q are distinct prime numbers, there are no lattice points on ON except for the endpoints O and N .

(b) The y -coordinate of M is

$$\frac{q}{p} \cdot \frac{p-1}{2} = \frac{q}{2} - \frac{1}{2} \frac{q}{p}.$$

(c) The y co-ordinate of M in (b) lies (5)
between two consecutive integers $\frac{q-1}{2}$ and $\frac{q+1}{2}$,

Since $\frac{q-1}{2} = \frac{q}{2} - \frac{1}{2} < \frac{q}{2} - \frac{1}{2} \frac{q}{p}$ (since $p > q$)
 $< \frac{q}{2}$
 $< \frac{q+1}{2}$.

Thus there are no lattice points in the triangle MPB, except possibly on PB.

Thus the number of lattice points in the rectangle OABC including those points on AB and BC, but not those points on OA and OC, is the number of lattice points in the triangle OPC, not including those points on OC, plus the number of lattice points in the triangle OAM, not including those points on OA. The number of lattice points

In the triangle OPC , not including those points on OC , is (6)

$$N_1 = \sum_{j=1}^{(q-1)/2} \left[\frac{jP}{q} \right].$$

The number of lattice points in the triangle OAM , not including those points on OA , is

$$N_2 = \sum_{j=1}^{(p-1)/2} \left[\frac{jQ}{p} \right].$$

Thus

$$\begin{aligned} N_1 + N_2 &= \sum_{j=1}^{(q-1)/2} \left[\frac{jP}{q} \right] + \sum_{j=1}^{(p-1)/2} \left[\frac{jQ}{p} \right] \\ &= \underbrace{\frac{(p-1)}{2} \cdot \frac{(q-1)}{2}}_{\text{by our first Count.}} \quad (5) \end{aligned}$$

Then by Lemma 4.10 we have that

$$\begin{aligned} \left(\frac{P}{q} \right) \left(\frac{Q}{p} \right) &= (-1)^{N_1} (-1)^{N_2} \\ &= (-1)^{N_1 + N_2} \\ &= (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \end{aligned}$$

by (5).

Example: Characterise the primes p (7)
(in terms of congruence conditions) for which
 11 is a quadratic residue modulo p .

We want all the primes such that $\left(\frac{11}{p}\right) = 1$.

Using quadratic reciprocity we have that

$$\left(\frac{11}{p}\right) = \begin{cases} \left(\frac{p}{11}\right) & \text{if } p \equiv 1 \pmod{4} \\ -\left(\frac{p}{11}\right) & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

Quadratic residues mod 11: $1, 3, 4, 5, 9 \pmod{11}$

Quadratic non-residues mod 11: $2, 6, 7, 8, 10 \pmod{11}$

Then use the Chinese Remainder Theorem
to solve each of the 10 pairs of congruences.

$$\begin{cases} p \equiv 1 \pmod{4} \\ p \equiv 1 \pmod{11} \end{cases} \xRightarrow{\text{Exercise}} p \equiv 1 \pmod{44}$$

$$\begin{cases} p \equiv 1 \pmod{4} \\ p \equiv 3 \pmod{11} \end{cases} \Rightarrow p \equiv 25 \pmod{44}$$

$$\begin{cases} p \equiv 1 \pmod{4} \\ p \equiv 4 \pmod{11} \end{cases} \Rightarrow p \equiv 37 \pmod{44}$$

$$\begin{cases} p \equiv 1 \pmod{4} \\ p \equiv 5 \pmod{11} \end{cases} \Rightarrow p \equiv 5 \pmod{44} \quad (8)$$

$$\begin{cases} p \equiv 1 \pmod{4} \\ p \equiv 9 \pmod{11} \end{cases} \Rightarrow p \equiv 9 \pmod{44}$$

$$\begin{cases} p \equiv 3 \pmod{4} \\ p \equiv 2 \pmod{11} \end{cases} \Rightarrow p \equiv 35 \pmod{44}$$

$$\begin{cases} p \equiv 3 \pmod{4} \\ p \equiv 6 \pmod{11} \end{cases} \Rightarrow p \equiv 39 \pmod{44}$$

$$\begin{cases} p \equiv 3 \pmod{4} \\ p \equiv 7 \pmod{11} \end{cases} \Rightarrow p \equiv 7 \pmod{44}$$

$$\begin{cases} p \equiv 3 \pmod{4} \\ p \equiv 8 \pmod{11} \end{cases} \Rightarrow p \equiv 19 \pmod{44}$$

$$\begin{cases} p \equiv 3 \pmod{4} \\ p \equiv 10 \pmod{11} \end{cases} \Rightarrow p \equiv 43 \pmod{44}$$

For primes p we have $\left(\frac{11}{p}\right) = 1$ if and only if $p \equiv 1, 5, 9, 19, 25, 35, 37, 39, 43 \pmod{44}$.

⑨

Example (Exercise): Characterise the primes p for which 5 is a quadratic residue modulo p .